CONTENTS

CHAPTER 1 Introductory Concepts 15

1-1 Digital and Analog Quantities 16
1-2 Binary Digits, Logic Levels, and Digital Waveforms 19
1-3 Basic Logic Functions 25
1-4 Combinational and Sequential Logic Functions 27
1-5 Introduction to Programmable Logic 34
1-6 Fixed-Function Logic Devices 40
1-7 Test and Measurement Instruments 43
1-8 Introduction to Troubleshooting 54

CHAPTER 2 Number Systems, Operations, and Codes 65

2-1 Decimal Numbers 66
2-2 Binary Numbers 67
2-3 Decimal-to-Binary Conversion 71
2-4 Binary Arithmetic 74
2-5 Complements of Binary Numbers 77
2-6 Signed Numbers 79
2-7 Arithmetic Operations with Signed Numbers 85
2-8 Hexadecimal Numbers 92
2-9 Octal Numbers 98
2-10 Binary Coded Decimal (BCD) 100
2-11 Digital Codes 104
2-12 Error Codes 109

CHAPTER 3 Logic Gates 125

3-1 The Inverter 126
3-2 The AND Gate 129
3-3 The OR Gate 136
3-4 The NAND Gate 140
3-5 The NOR Gate 145
3-6 The Exclusive-OR and Exclusive-NOR Gates 149
3-7 Programmable Logic 153
3-8 Fixed-Function Logic Gates 160
3-9 Troubleshooting 170

CHAPTER 4 Boolean Algebra and Logic Simplification 191

4-1 Boolean Operations and Expressions 192
4-2 Laws and Rules of Boolean Algebra 193
4-3 DeMorgan's Theorems 199

4-4 Boolean Analysis of Logic Circuits 203
4-5 Logic Simplification Using Boolean Algebra 205
4-6 Standard Forms of Boolean Expressions 209
4-7 Boolean Expressions and Truth Tables 216
4-8 The Karnaugh Map 219
4-9 Karnaugh Map SOP Minimization 222
4-10 Karnaugh Map POS Minimization 233
4-11 The Quine-McCluskey Method 237
4-12 Boolean Expressions with VHDL 240
Applied Logic 244

CHAPTER 5 Combinational Logic Analysis 261
5-1 Basic Combinational Logic Circuits 262
5-2 Implementing Combinational Logic 267
5-3 The Universal Property of NAND and NOR gates 272
5-4 Combinational Logic Using NAND and NOR Gates 274
5-5 Pulse Waveform Operation 279
5-6 Combinational Logic with VHDL 283
5-7 Troubleshooting 288
Applied Logic 294

CHAPTER 6 Functions of Combinational Logic 313
6-1 Half and Full Adders 314
6-2 Parallel Binary Adders 317
6-3 Ripple Carry and Look-Ahead Carry Adders 324
6-4 Comparators 327
6-5 Decoders 331
6-6 Encoders 341
6-7 Code Converters 345
6-8 Multiplexers (Data Selectors) 347
6-9 Demultiplexers 356
6-10 Parity Generators/Checkers 358
6-11 Troubleshooting 362
Applied Logic 365

CHAPTER 7 Latches, Flip-Flops, and Timers 387
7-1 Latches 388
7-2 Flip-Flops 395
7-3 Flip-Flop Operating Characteristics 406
7-4 Flip-Flop Applications 409
7-5 One-Shots 414
7-6 The Astable Multivibrator 423
7-7 Troubleshooting 427
Applied Logic 429

CHAPTER 8 Shift Registers 449

8-1 Shift Register Operations 450
8-2 Types of Shift Register Data I/Os 451
8-3 Bidirectional Shift Registers 462
8-4 Shift Register Counters 465
8-5 Shift Register Applications 469
8-6 Logic Symbols with Dependency Notation 476
8-7 Troubleshooting 478
Applied Logic 480

CHAPTER 9 Counters 497

9-1 Finite State Machines 498
9-2 Asynchronous Counters 500
9-3 Synchronous Counters 507
9-4 Up/Down Synchronous Counters 515
9-5 Design of Synchronous Counters 519
9-6 Cascaded Counters 527
9-7 Counter Decoding 531
9-8 Counter Applications 534
9-9 Logic Symbols with Dependency Notation 539
9-10 Troubleshooting 541
Applied Logic 545

CHAPTER 10 Programmable Logic 561

10-1 Simple Programmable Logic Devices (SPLDs) 562
10-2 Complex Programmable Logic Devices (CPLDs) 567
10-3 Macrocell Modes 574
10-4 Field-Programmable Gate Arrays (FPGAs) 577
10-5 Programmable Logic software 585
10-6 Boundary Scan Logic 595
10-7 Troubleshooting 602
Applied Logic 608

CHAPTER 11 Data Storage 627

11-1 Semiconductor Memory Basics 628
11-2 The Random-Access Memory (RAM) 633
11-3 The Read-Only Memory (ROM) 646
11-4 Programmable ROMs 652
11-5 The Flash Memory 655
11-6 Memory Expansion 660
11-7 Special Types of Memories 666
11-8 Magnetic and Optical Storage 670
11-9 Memory Hierarchy 676
11-10 Cloud Storage 680
11-11 Troubleshooting 683

CHAPTER 12 Signal Conversion and Processing 697

12-1 Analog-to-Digital Conversion 698
12-2 Methods of Analog-to-Digital Conversion 704
12-3 Methods of Digital-to-Analog Conversion 715
12-4 Digital Signal Processing 723
12-5 The Digital Signal Processor (DSP) 724

CHAPTER 13 Data Transmission 739

13-1 Data Transmission Media 740
13-2 Methods and Modes of Data Transmission 745
13-3 Modulation of Analog Signals with Digital Data 750
13-4 Modulation of Digital Signals with Analog Data 753
13-5 Multiplexing and Demultiplexing 759
13-6 Bus Basics 764
13-7 Parallel Buses 769
13-8 The Universal Serial Bus (USB) 775
13-9 Other Serial Buses 778
13-10 Bus Interfacing 784

CHAPTER 14 Data Processing and Control 801

14-1 The Computer System 802
14-2 Practical Computer System Considerations 806
14-3 The Processor: Basic Operation 812
14-4 The Processor: Addressing Modes 817
14-5 The Processor: Special Operations 823
14-6 Operating Systems and Hardware 828
14-7 Programming 831
14-8 Microcontrollers and Embedded Systems 838
14-9 System on Chip (SoC) 844

ON WEBSITE: http://www.pearsonglobaleditions.com/floyd
CHAPTER 15 Integrated Circuit Technologies 855

15-1 Basic Operational Characteristics and Parameters 856
15-2 CMOS Circuits 863
15-3 TTL (Bipolar) Circuits 868
15-4 Practical Considerations in the Use of TTL 873
15-5 Comparison of CMOS and TTL Performance 880
15-6 Emitter-Coupled Logic (ECL) Circuits 881
15-7 PMOS, NMOS, and E^2CMOS 883

ANSWERS TO ODD-NUMBERED PROBLEMS A-1

GLOSSARY A-31

INDEX A-42